Product Case Solving: Increasing Swiggy Instamart's Average Order Value

SWIGGY instamart

Deck Overlook

1. Business and Product Outcome Mapping
2. Market Understanding and User Segmentation
3. User Research and Root Cause Identification
4. Problem Framing and Impact Mapping
5. Solution Ideation and Prioritization
6. Prototyping and Solution Detailing
7. System Design, Constraints and Second Order Thinking
8. Metrics & Pitfall Mitigation



The Quick Commerce market in India

Forecasted to generate a revenue of ₹271.80bn in 2024.

Expected annual growth rate (2024-2029) of 25.67%

The user penetration rate, which currently stands at 1.8% in 2024, is projected to rise to 4.0% by 2029.

Swiggy Instamart is a quick commerce service offering 10-minute deliveries for a wide range of products.

Actors in the Eco system

1. Customer
2. Instamart App
3. Delivery Partners

4. Warehouse Employees
5. Relatives/ Friends

[secondary actors]

Product Outcomes

- Average order value (AOV)
  - Total value of all orders
  - /by Total no. of orders
- x Total Number of orders
- Revenue
  - + Delivery fee
  - +Advertisement/partnership payments
  - + Miscellaneous

Business Outcome

Profitability through Instamart

- - Cost
  - -Operation cost [App+ Employee Salary]
  - -Delivery cost
  - -Cost of goods
  - -Warehouse Rent
  - -Packaging Cost
  - -Discounts
  - -Miscellaneous

Why is AOV Important?

With an increase in avg. order value, unit economics becomes profitable (as shown in the calculations below) which helps in achieving business outcomes.

Rationale behind Product Outcomes:

- **Decreasing order numbers:** Combining orders, as long as the total value of orders remains consistent or increases, can help optimize operations and improve profitability.
- **Increasing the total value of orders:** Total order value is directly proportional to the average order value. Ensuring each order contributes to overall revenue goals.

But how do we reach profitability?

- **Desired Profit per Order:** INR 10 **[Assumption]**

Target Revenue per Order: To reach a profit of INR 10 per order:

- Target Revenue per Order = Cost per Order + Desired Profit
- Target Revenue per Order = INR 465 + INR 10 = INR 475

E.I. AOV needs to be increased by 10% to reach 440 rs.

Final Objective = Increase AOV by 10% to reach profitability through Instamart.

Business Outcome: Achieving Profitable Unit Economics

Let's Understand the current unit economics

Revenue Items

- **Average Order Value:** INR 400 (Based on industry reports and average basket size)
- **Number of Orders per Month:** 450,000 (Based on industry reports)
- **Delivery Fee:** INR 20 (Standard delivery fee in most cities)
- **Advertisement/Partnership Payments:** INR 10 (Based on typical marketing and partnership agreements)
- **Miscellaneous Revenue:** INR 5 (Includes potential add-ons or promotions)

- **Revenue per Order = Average Order Value + Delivery Fee + Advertisement/Partnership Payments + Miscellaneous Revenue**
- **Revenue per Order = INR 400 + INR 20 + INR 10 + INR 5 = INR 435**

Profit Calculation:

Profit per Order = Revenue per Order - Cost per Order

Profit per Order = INR 435 - INR 465 = -INR 30

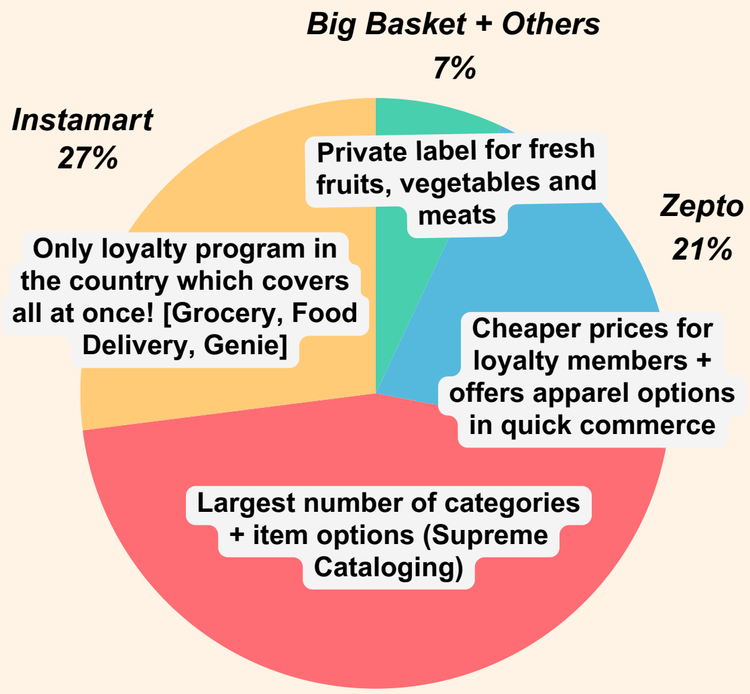
i.e Rs 30 loss per order [Approximately]

Cost Items

- **System Cost (App & Salary):** INR 30 (Based on platform maintenance and employee costs)
- **Delivery Cost:** INR 40 (Includes driver wages)
- **Cost of Goods:** INR 360 (Assumes a 10% margin on average, considering vendor costs and markups)
- **Warehouse Rent:** INR 15 (Based on warehouse size and operational costs)
- **Packaging Cost:** INR 5 (Assumes minimal packaging for grocery deliveries)
- **Discounts:** INR 10 (Assumes a discount rate of 3% on average, including promo codes)
- **Miscellaneous Cost:** INR 5 (Includes potential operational costs and miscellaneous expenses)

- **Cost per Order = System Cost + Delivery Cost + Cost of Goods + Warehouse Rent/Capex + Packaging Cost + Discounts + Miscellaneous Cost**
- **Cost per Order = INR 30 + INR 40 + INR 360 + INR 15 + INR 5 + INR 10 + INR 5 = INR 465**

Objective: To increase average order value by 10% to reach profitability through swiggy instamart.



MARKET SHARE ACROSS QUICK COMMERCE APPS IN INDIA IN 2024 AND COMPETITIVE DIFFERENTIATION

User Segmentation

(Based on behavioural tendencies)

Order Frequency	Monetization
POWER USERS [18% USERS] Users with 15+ orders in a month	**HIGH AVERAGE ORDER VALUE [13% USERS]** More than 800 Rs
MODERATE USERS [43% USERS] 6-15 orders in a month	**MODERATE AVERAGE ORDER VALUE [12% USERS** 401-800 Rs
CASUAL USERS [39% USERS] 5 or less orders in a month	**LOW AVERAGE ORDER VALUE [75% USERS]** Less than or equal to 400 Rs
	FINAL SEGMENT [50% USERS] (AOV ≤ INR 400 and order frequency > 5 times a month)

Rejected segment as the number of orders is already low, hence opportunity size diminishes

Rejected segment as the average order value is already on a higher side, thus leaves low propensity for increase in order value.

User Research:

A user survey was conducted with 40 participants from Tier 1 city, aged between 18-24 years to identify the segments and the Pain points:

75% of users have an average order value of less than 400Rs! ✓ **H1 Accepted**

39% of users order 5 or less times in a month!

13% of users have an average order value of more than 800Rs! ✓ **H2 Accepted**

18% of users order more than 15 times a month!

Problem Hypothesis:

H1: Majority of quick commerce users [50%+1] will have an average order value of less than 400 Rs.

H2: Less than 15% quick commerce users will have an average order value of more than 800 Rs.

H3: Majority of quick commerce users order multiple low value orders based on urgency because of lack of pantry management.

Target User Segment: Users with average order value equal to or over INR 400 with an order frequency of more than 5 times a month.

Problem breakdown: Urgency- Based Ordering and Price Comparison

User Insight
Urgency-Based Ordering:
Most users order impulsively when they run out of something, not planning ahead or making larger, stock-up purchases. This leads to frequent, small orders.

Why does this happen?
Lack of Pantry Management:
Many users don't keep track of their inventory, leading to last-minute orders and missing opportunities for bulk discounts.

This leads to multiple fragmented orders which leads to low order value.
(Root Cause of Low AOV)

User Insight
Price Comparison:
Users feel the need to compare prices across apps and offline stores. They might find better deals elsewhere or perceive the app's prices to be generally high.

Why does this happen?
Price sensitivity:
Users are sensitive to price and perception of high prices due to lack of comparison makes them hesitant to commit to larger orders.

Ideal User Persona



Vaibhav Sharma

Age: 23
Occupation: Data Scientist
Location: Delhi
Lives: with family
Income Range: 55,000-60,000 Rs
Tech Savvy

"I wish someone could just take care of my pantry and let me know before I run out of groceries!"

FRUSTRATIONS/ PAIN POINTS

- Limited motivation for keeping a check on pantry items.
- Hectic schedule leading to a lack of organization.

GOALS

- To be able to manage pantry without rushing into last minute orders every day.

MOTIVATIONS

- Comfort, Convenience, Price

"I usually order when I need something very urgently just enough to avail the free delivery."
- 24 Yr old, IT Worker

"My work is quite Hectic and doesn't allow me a lot of time to make a consolidated list of things I need. I just order as and when urgency arises."
-23 Yr old, Data Scientist

"I compare prices across apps and order just enough to get by the day. I am constantly checking when I will surpass the free delivery line."
-23 Yr old, Software Developer

H3 Accepted

Root cause of missed business outcome : Multiple fragmented orders throughout the days hinder business outcome by reducing average order value and decreasing profitability.

Problem Framing

What is the value generated for the Target Segment by solving this problem?

Solving this problem for the target segment, comprising 50% of users, offers significant benefits. Firstly, it alleviates the stress of remembering what to order and when, leading to a more relaxed and organized lifestyle. Additionally, it enables them to avail bulk discounts by consolidating their orders, resulting in potential cost savings

Why should we solve this problem now?

The urgency to address this problem is evident due to India's demographic landscape, with over 50% of the population below the age of 25 and more than 65% below the age of 35. **The majority of this population segment heavily relies on technology for various tasks.** However, traditional pantry management practices, typically handled by relatives or parents through offline mediums, have left this demographic with **outdated methods.** Given the **ongoing expansion of the mobile market and the inevitable penetration of technology** into various aspects of daily life, addressing pantry management for youth now is crucial to ensure they are not left behind and to meet their evolving needs effectively.

How do we know it's a real problem?

The problem is validated through user survey data, which reveals that 50% of users place orders more than six times a month, yet maintain an average order value of less than 400 Rs. This discrepancy indicates a pattern of frequent but small-value orders, suggesting a genuine need for a solution to streamline ordering processes and potentially increase order value.

What is the true problem?

The true problem lies in the lack of inventory management among users, resulting in last-minute orders and missed opportunities for bulk discounts. This contributes to a perception of higher prices, as users are unable to take advantage of bulk discount offers.

Jobs to be done

When I attempt to purchase affordable groceries online,
but struggle due to ordering based on urgency and inability to recall all necessary items,
which leads to rushed orders and missed bulk discounts.
Help me organize my pantry and grocery needs
so that I can avoid daily rushes for orders while taking advantage of bulk discounts.

Who are the customers facing the problem?

Tier 1 youth aged 18-24 who lack the time and motivation to manage their pantry yet prioritize cheaper prices and convenience.

Problem Identification & Validation : Users struggle to purchase large quantity of groceries online, due to an urgency-based order behavior and lack of pantry management, leading to missed bulk discounts.

Impact Mapping

What value is generated by solving this for the business?

The business is able to increase the Average Order Value (AOV) by 10% and achieve profitability.

Let's Understand this through **Impact mapping.**

If we encourage users in the target segment to consolidate their orders, then,

New Number of Average Orders = Original Average Number of Orders/ 1+Percentage Increase in Order Value

Percentage Decrease in Number of orders=(Original Number of Orders−New Number of Orders/ Original Number of Orders)×100

Overall AOV Increase for Target Segment=Percentage Increase in AOV × Proportion of Target Segment Users Who Consolidate Orders

Overall AOV Increase=Overall AOV Increase for Target Segment × Proportion of Target Segment in Total User Base

New AOV=Current AOV+(Current AOV×Overall AOV Increase)

Let's put in numbers to justify our approach.

Target Segment: 50% of total users.

Current Average Order Frequency of the target segment: 12 orders per month.

Assumptions

50% of target segment users will consolidate their orders.

These users will consolidate 40% of their orders.

Current AOV: 400 Rs.

New Number of Orders = 12/1.4 = 8.57 orders [Approx

Percentage Decrease in Number of orders = (12-8.57/12) ×100

Overall AOV Increase for Target Segment = 40% × 50% = 20%

Overall AOV Increase= 20% × 50% = 10%

New AOV = 400 + (400 × 10%) = 440 Rs

This is the business value it creates! The business is able to increase the Average Order Value (AOV) by 10% and achieve profitability!

Impact Mapping: By addressing the identified problem, we can increase average order value by 10% to reach profitability.

Solution Ideation: InstaRestocker and InstaPrice Comparer

InstaRestocker

Objective: Enhance customer convenience and retention by implementing an AI-based restocking tool that predicts when users will need to restock groceries based on their personal usage patterns, for both perishable and non-perishable items.

If we implement an AI-based restocking tool that predicts when users will need to restock groceries based on their personal usage patterns for both perishable and non-perishable items, then we will enhance customer convenience and retention because it will ensure users never run out of essential items and they don't have to manage their pantry on their own.

InstaPrice Comparer

Objective: Enhance customer price perception by implementing a price comparer tool that compares prices across quick commerce apps and offline stores for high value items based on market data, shifting perceptions about Instamart price ranges.

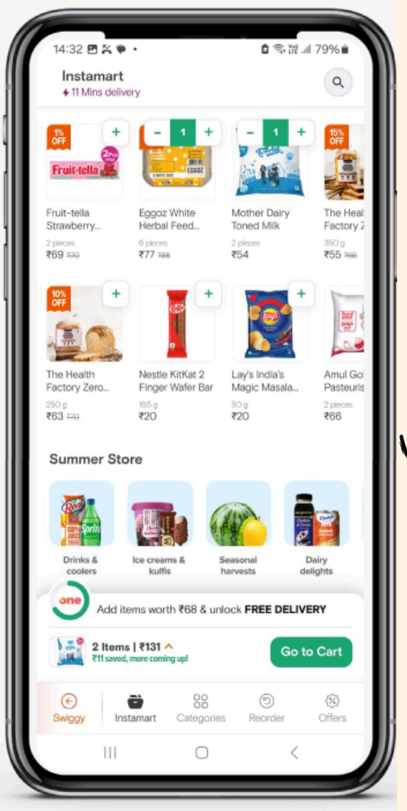
If we implement a price comparer tool that compares prices across quick commerce apps and offline stores for high-value items based on market data, then we will enhance customer price perception and increase order value because it will provide assurance that they are not overpaying.

Product Solution validation is based on - Value, Usability, Feasibility, Viability, and Scalability

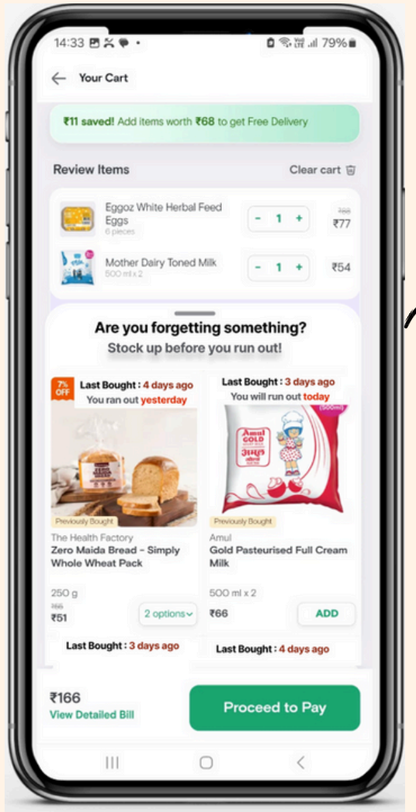
Solution Prioritization

Product score elements	InstaRestocker	InstaPrice Comparer
Impact	9	5
Confidence	.8	.5
Effort	5	4
Product Score = [Impact x Confidence] - Effort	2.2	-1.5

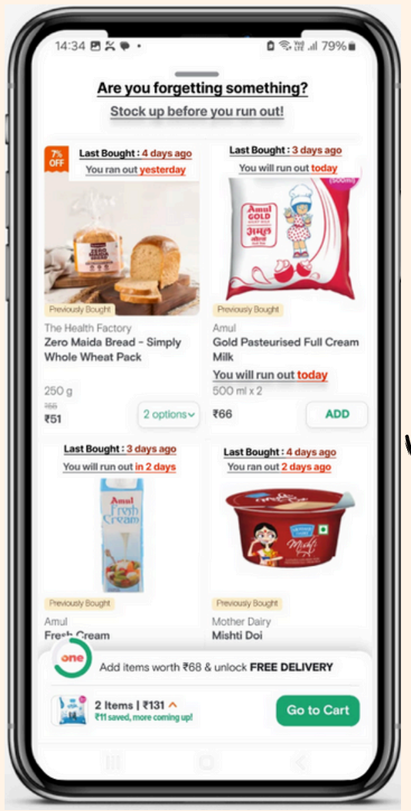
Solution ideation & Prioritization: Out of the two proposed solutions, the InstaRestocker is prioritized for implementation.



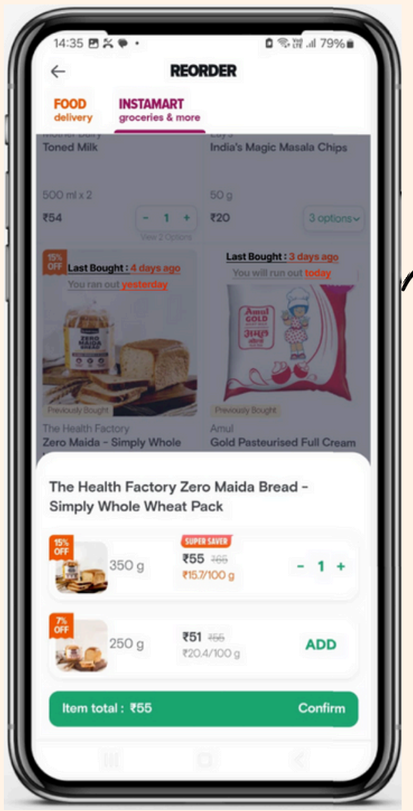
User visits the app, adds the items, and goes to cart



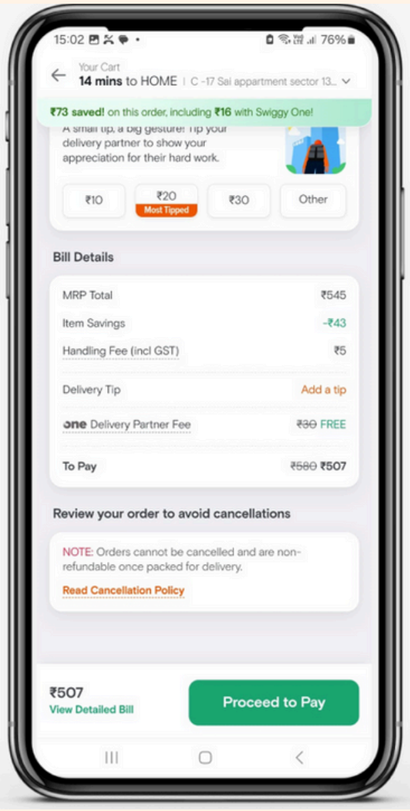
On the cart page they see the 'InstaRestocker' card which nudges them to stock up before they run out of products



They can scroll up to make it an immersive restocking experience and to view all products. [Helps in activation]



The same feature is also available at the reorder page [Recall + Engagement]



Once they are done with adding all the items, they can check out.

Prototyping and Detailing

Perishable Items Management

Data Display: Display expiry date, last purchase date, and estimated depletion date based on user usage patterns.

Algorithmic Prioritization:

1.Automatically determine the order in which items are displayed.

2. Items with a shorter shelf life (1-10 days) are shown 1-3 days before expected depletion.

3. For items with longer shelf life, recommend restocking within a timeframe based on usage patterns or expiry/best before date.

Non-Perishable Items Management

Lifespan Tracking:

1.Record typical item lifespan and usage frequency.

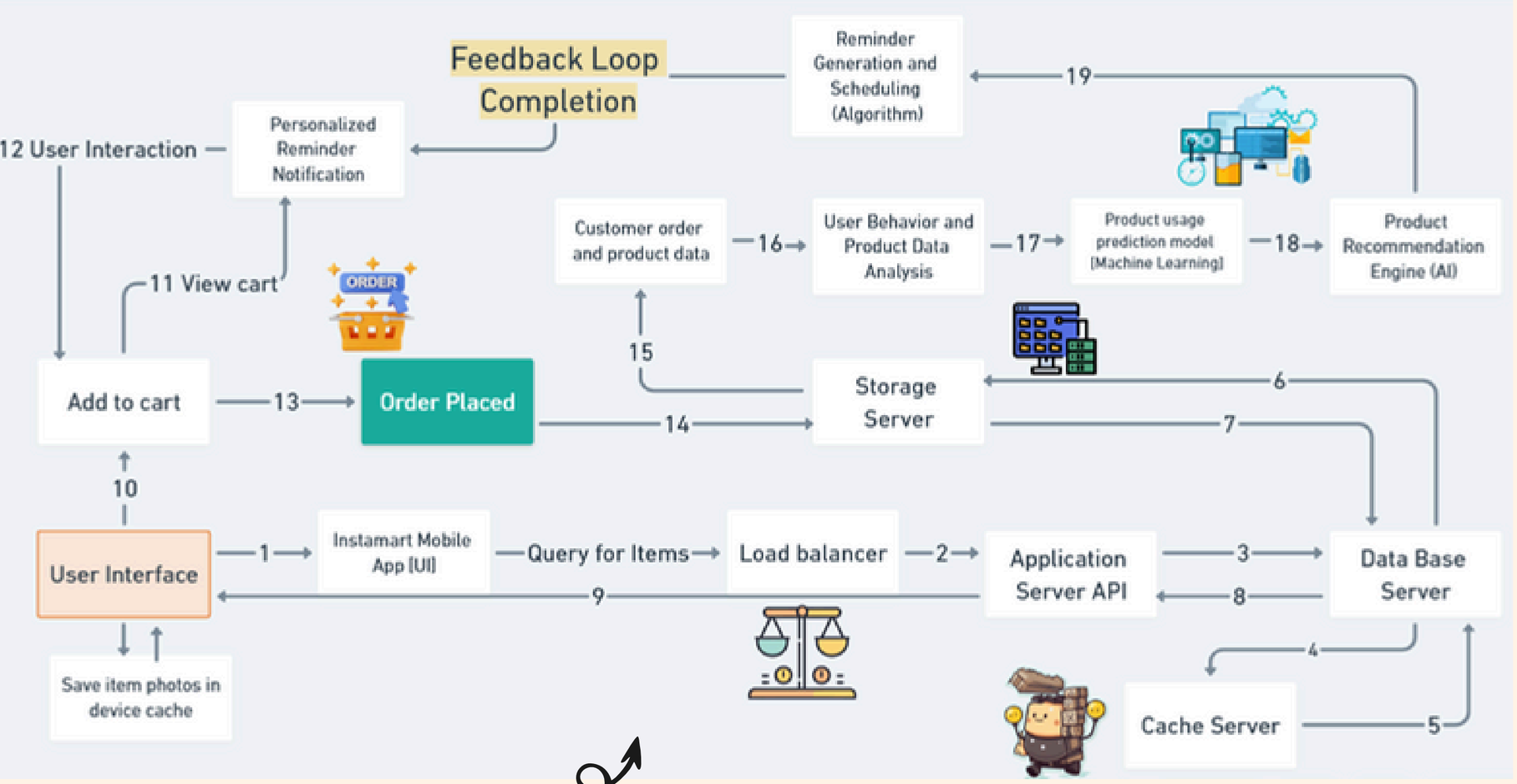
2. Recommend restocking based on expected end-of-life or depletion.

Alert Information:

Display item name, last purchase date, typical lifespan.

Recommend restocking with a message like "You might need to buy it again in a week," tailored to usage patterns.

Detailed Solutioning and Prototyping: InstaRestocker suggests items to add to your cart during checkout, based on your past usage, to help you avoid running out of essentials.



Constraints

1. **Post-Depletion Handling:**

Keep items on the list for a week after depletion.

Provide context-specific notifications like "You ran out 2 days ago".

Update algorithm periodically for improved usage understanding.

2. **Ethical Considerations**: Respect user autonomy and preferences in decision-making regarding data tracking and usage.

Second Order Thinking

1. A decrease in order numbers could significantly impact delivery personnel's compensation, necessitating a review of the compensation structure and incentives.
2. Additionally, higher order values may result in constant travel with heavy or expensive items, posing security and safety concerns that require careful consideration and planning.

System Design

Hypothesis for the chosen solution:

For who

Change

Opening up a section at the cart that shows when users, who make multiple small average order value orders, will run out of products, in their pantry will nudge them to consolidate orders and will increase average order value by 10% over a period of 3 months.

Impact

Measure

When

Measuring Success

Metric Type	Metric
North star	- %Increase in Profitability as a function of AOV
L1	- % Increase in Average order value
L2	- % Increase in total order value - % Decrease in order numbers [Given total order value does not decrease]
Activation	- Click through rate for enabling the full-screen mode for 'InstaRestock' - % Users that visit the reorder section
Adoption	-% of users that add 1 item from 'InstaRestock' - Average time spent by a user in the first session of exploration.
Engagement	- Avg time spent on 'InstaRestock' per user session -% of items added in one single order from 'InstaRestock' - % of users ordering primarily from 'InstaRestock'
Retention	% of repeat users revisiting after a week
Customer Satisfaction	NPS(Net Promoter Score) to know how likely the present activated users are likely to recommend the feature to family and friends.

Possible Pitfalls

Risk: The algorithm's accuracy may be compromised if users make purchases from sources other than Swiggy, potentially leading to tracking errors and incorrect predictions.
Opportunity: Implement cross-platform integration to accurately track user purchases from various sources, ensuring data consistency and precise predictions within the algorithm.

Risk: The absence of nudges for free delivery due to already high order values could result in reduced purchasing behavior, with users opting for need-based items rather than splurging.
Opportunity: Incorporate targeted nudges or incentives beyond free delivery to encourage additional purchases and prevent a shift towards strictly need-based shopping.

Risk: There's a risk that users may exclusively purchase the same products repeatedly, resulting in the cannibalization of need-based purchases and a limited exploration of new products, potentially missing out on higher-value offerings.
Opportunity: In later stages of product, a 'try this instead' nudge could be introduced to combat this.

Success Metrics & Potential Pitfalls: Metrics track product success, user satisfaction and, engagement. While pitfalls focus on data accuracy, meeting diverse needs, and user-friendliness.